

Shashi Bhushan Vijay

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SUMMARY

M.Tech Computer Science Engineering student and patent holder specializing in ML pipelines, distributed systems, and backend engineering. Proven track record of shipping production-grade systems - including a patented indoor positioning solution with $R^2 = 0.978$ accuracy. Experienced across the full stack, from edge hardware integration to LLM-powered SaaS platforms.

EXPERIENCE & PROJECTS

AutoML Assistant

Jan 2026

Project

Repo | Live Demo

- Architected an end-to-end AutoML platform using Streamlit + Python to automate data upload, preprocessing, analysis, model training, and optimization workflows.
- Designed a modular ML backend supporting **regression, classification, and time-series forecasting** with algorithms like XGBoost, Random Forest, ARIMA/SARIMAX, and optional LSTM.
- Integrated LLM-powered model recommendation (RAG) using LangChain, Groq (Llama 3.1), and FAISS to suggest suitable models based on dataset statistics and ML best practices.
- Implemented **SHAP-based explainability** (feature importance, beeswarm, waterfall, dependence plots) to make model predictions interpretable for end users.
- Exposed trained models through a **FastAPI** inference API (/predict, /predict/csv) with **persistent workspace/model management** for reusable deployment-ready ML pipelines.

Patent: Coordinated Indoor Position Determination via Multi-Stage Signal Processing Dec 2024–Dec 2025

VIT, Vellore, IN

Application No. 202541115892

- Delivered an indoor localization system using Wi-Fi RSSI fingerprinting and Bluetooth integration, achieving approx. **1.6 m positioning accuracy (MAE: 1.468 m, R^2 : 0.978)** in complex environments.
- Deployed an ESP32-based autonomous Wi-Fi sniffer on a robotic car, automating radio-map generation and capturing **4,277 RSSI records across 209 access points** - eliminating manual site surveys.
- Designed and optimized a Dynamic Stacking ensemble (KNN, Random Forest, XGBoost + Gradient Boosting meta-learner), cutting localization error by **approx. 43% (RMSE: 3.124 to 1.787)**.
- Engineered a real-time data pipeline using MQTT, Raspberry Pi, and Python, transforming raw Wi-Fi scans into machine-learning-ready datasets for live location predictions.
- Implemented an **A*-based** indoor navigation module using NetworkX for optimal route generation and dynamic path recalculation from predicted user locations to destinations.

OncoFollow - Full-Stack Oncology Follow-Up Platform

Apr 2026 – Jun 2026

Project

Repo | Live Demo

- Launched OncoFollow, a full-stack clinical workflow platform in **Next.js 14 and React 18** unifying patient symptom reporting, clinician alert queues, admin oversight, and longitudinal care timelines - reducing triage overhead and improving response time to high-acuity cases.
- Designed and implemented a **serverless API layer** (Next.js API routes) proxying Groq Cloud LLM inference with role-specific prompt controls, server-side secret management, and structured JSON contracts - applying production-grade **cloud architecture** and **API design** patterns.
- Engineered a rule-based triage engine across **20 symptom categories** and 4 scoring dimensions (severity, duration, frequency, trend); incorporated red-flag detection and override logic, achieving **99% sensitivity** for emergency-case interception.
- Crafted real-time **analytics dashboards** (Recharts) with lab-trend sparklines, NEWS2 acuity scoring, and AI-powered trend analysis for actionable clinical insights.
- Anchored all AI outputs in NCCN/ASCO guidelines with citation-backed responses ([R1], [G1]); incorporated audit logging, consent management, and blocked-phrases filtering for **HIPAA-adjacent compliance**.

FLEX-DCA AI Platform - AI-Powered Debt Collection Allocation & Compliance System

Jan-Feb 2026

Project

Repo

- Engineered a full-stack debt recovery platform using FastAPI, React, and SQLite to ingest cases, run predictions, auto-allocate DCAs, and track operational KPIs via REST APIs.
- Trained and integrated **3 CatBoost models** (recovery probability, aging risk, closure speed) with financial feature engineering for real-time decision support.
- Implemented a **RAG-based** legal copilot using **FAISS + SentenceTransformers + Groq LLaMA 3.1**, enabling policy-grounded answers for FDCPA, Regulation F, and RBI compliance queries.
- Created an operations dashboard with KPI monitoring and case workflows, improving visibility into recovered/open cases and allocation outcomes.
- Structured backend into modular services (ML, allocation, RAG, dashboard, ingestion) to support scalable, production-style architecture for enterprise debt operations.

EDUCATION

Vellore Institute of Technology , Vellore	<i>Expected April 2027</i>
Integrated M.Tech in Computer Science Engineering	GPA: 7.0/10
<i>Relevant Coursework: Machine Learning, Distributed Systems, Data Structures & Algorithms, Database Management, Computer Networks, Operating Systems, Cloud Computing</i>	
SM Public School , Haridwar	<i>2020–21</i>
Junior College (Class XII)	91.2%
SM Public School , Haridwar	<i>2018–19</i>
High School (Class X)	87%

TECHNICAL SKILLS

Languages & Core: Python, C++, Java, JavaScript, SQL
Backend & Systems: FastAPI, REST APIs, Microservices, MQTT, Docker, CI/CD
Databases: PostgreSQL, SQLite, Redis | **Tools & Cloud:** Git, AWS, Streamlit
ML & AI: Scikit-learn, XGBoost, TensorFlow, PyTorch, LSTM, SHAP, RAG, FAISS, LangChain, Groq

LEADERSHIP & EXTRACURRICULARS

- Sponsorship Coordinator**, ISTE Chapter – VIT Vellore
- Secured **Rs. 1,00,000+ in corporate sponsorships** for flagship hackathon (Horizon) by pitching to **100+ founders and startups**; orchestrated end-to-end event logistics and developed outreach frameworks that established strategic industry connections for future initiatives.

ACHIEVEMENTS

1st Place, ARC Hackathon (ARC Lab, VIT) – Created a GUI-based AutoML tool that automatically benchmarks and recommends the best ML model for any given dataset, winning against competing teams.

CERTIFICATIONS

Stock Market Basics & Technical Analysis – View Certificate
Designing Seamless Notification Service – View Certificate